antropy: a dimension-agnostic entropy methods framework

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Computational methods manuscript accompanying antropy-hd 0.1.0. Generated from the repository's deterministic experiment plan; this is an implementation report.

Abstract

antropy organizes high-dimensional sample entropy and related complexity estimators around validated spatial embedding, explicit estimator conventions, typed results and reproducible orchestration. Local templates preserve native coordinate structure before each window is flattened. Probability, ordinal, spectral and trajectory measures share explicit finite, undefined and infinite statuses. A seeded synthetic demonstration covers the configured spatial dimensions and fields at several tolerances. The software and numerical examples are intended to support reproducible methods development; they do not reproduce the empirical experiments in Xia and Wang (2026), establish new scientific findings, or define an estimator-independent ordering of complexity.

Motivation and scope

Spatial data should not be flattened globally before local templates are formed. A dimension-agnostic framework makes geometry, delay, stride, tolerance scaling, exclusions, distance and memory budgets visible. The repository name combines entropy and ant, reflecting interest in collective biological systems. The distribution antropy-hd and import antropy_hd avoid shadowing the separate AntroPy time-series library. The present examples use generated arrays only.

HDSE and local template estimators

HDSE follows Xia and Wang's high-dimensional sample entropy construction: compare m-sized local blocks and their m+1 extensions over common valid origins, use maximum absolute component distance and a strict threshold r times global standard deviation, exclude a template's comparison with itself, and return the negative natural logarithm of the ratio of extended to base matches. Population standard deviation (ddof=0) is the explicit default. The source's prose requires every coordinate to differ, while its equations exclude the identical origin tuple; these define different pair populations; ordinary tuple self-exclusion is the default, and coordinate exclusion is exposed as sensitivity configuration. The implementation contract records this decision.

Zero base matches yield undefined status; positive base matches and zero extended matches yield infinite status. Constant fields at zero strict tolerance produce no matches. Rectangular windows, selected extension axes, other norms, delay, stride and Theiler exclusions are declared extensions. One-dimensional sample entropy, approximate entropy using separate complete origin populations with self matches, fuzzy entropy with exp(-(distance/tolerance)^power), distance-distribution entropy, and trimmed block-mean multiscale analysis have separately documented contracts. Distance-distribution entropy uses all valid local origins; the original DistEn paper omits the final valid origin, so this is an explicitly declared extension. Fuzzy sums use log-domain accumulation to preserve small similarities. Multiscale analysis preserves original-data tolerance by default.

Related estimator definitions

For a normalized discrete distribution p, Shannon entropy is -sum(p log_b p), with zero-mass terms contributing zero. Renyi entropy is log_b(sum(p^q))/(1-q), with support-size, Shannon and maximum-probability limits at q=0, q=1 and positive infinity. Tsallis entropy is (1-sum(p^q))/(q-1), with k=1 and a natural-log Shannon limit at q=1. Joint entropy applies Shannon to the full probability tensor; conditional entropy is H(X,Y)-H(Y), and mutual information is H(X)+H(Y)-H(X,Y). Jensen-Shannon divergence is the weighted mixture entropy minus the weighted component entropies. These probability identities do not estimate a distribution from amplitude data by themselves.

Permutation entropy applies Shannon to local ordinal-pattern frequencies; weighted permutation entropy replaces counts by local variance weights. Spectral entropy applies Shannon to normalized FFT powers. SVD entropy applies Shannon to normalized singular values of the trajectory matrix. Normalization divides by the documented maximum for the probability alphabet, possible ordinal patterns, Fourier bins or singular-value count. Zero spectral energy and zero total ordinal variance are undefined, while unnormalized Shannon entropy for a singleton alphabet is zero. Distance-distribution entropy applies Shannon to a configured histogram of pairwise local-template distances. The histogram range and count affect that estimate.

Related estimators and interpretation

Discrete Shannon, Renyi, Tsallis, joint and conditional entropy, mutual information and Jensen-Shannon divergence consume normalized probability arrays. Histogram entropy explicitly discretizes observed amplitudes. Spatial permutation entropy uses stable position-based tie ordering within local windows; its weighted form weights by local population variance. N-dimensional spectral entropy normalizes the full FFT periodogram, retaining conjugate bins individually. SVD entropy normalizes singular values of the local trajectory matrix without squaring them. These conventions are not interchangeable: output units, normalization, alphabet, and input meaning are part of each result. A value from one estimator should not be treated as a calibrated value of another estimator.

Software architecture and verification

The numerical package depends on NumPy, Pydantic and Typer. Numerical routines do not import plotting, manuscript rendering or network services. Configurations reject unknown fields, method dispatch binds the full function signature, arrays must be finite real numeric data, and resource limits constrain template and pair work. Memory bounds cover controlled template and pair workspace, excluding caller arrays, parsing, SD temporaries and process RSS. Nonfinite eligible fuzzy distances or log weights raise a numerical-range error instead of returning a mathematical undefined estimate. The CLI reads NPY with pickle disabled and CSV as a two-dimensional grid. Strict JSON represents nonfinite estimates as null plus an explicit status. Tests use analytical examples, a direct loop oracle, seeded numerical inputs, real file boundaries and subprocesses. Deliberately malformed inputs and missing or altered artifacts must cause checks to fail. CI runs lint, strict type checks, branch-inclusive source coverage at a 90 percent floor, artifact validation and an installed-wheel CLI scenario. Passing these checks is evidence for the tested contracts, not proof of every possible scientific use.

Synthetic experiment design

The complete design is in experiment_plan.yaml. Independent seeded generators produce periodic fields, Gaussian noise and periodic fields with additive Gaussian noise. Fields have shapes 64, 16 x 16, 8 x 8 x 8. HDSE uses shape 2 on every axis and relative tolerances 0.2, 0.5, 1.0; default strict Chebyshev matching and self-exclusion are retained. Histogram entropy, normalized spatial permutation entropy and normalized full-FFT entropy are also recorded for each field. Changing dimension changes both window feature count and finite sample geometry; the experiment is a software demonstration rather than a controlled scientific comparison attributing differences solely to dimension.

Computational results

The pipeline recorded 54 estimator rows: 45 finite, 3 undefined and 6 infinite. Every status is retained. The following HDSE values are generated directly from the CSV. No monotonic relationship or classification accuracy is claimed.

Dimension	Structure	Relative radius r	HDSE (nats)	Status
1	periodic	0.2	0	finite
1	periodic	0.5	0.451985	finite
1	periodic	1.0	0.245482	finite
1	noise	0.2	1.4816	finite
1	noise	0.5	1.22536	finite
1	noise	1.0	0.680827	finite
1	mixed	0.2	2.70805	finite
1	mixed	0.5	1.02233	finite
1	mixed	1.0	0.488638	finite
2	periodic	0.2	0.262364	finite
2	periodic	0.5	1.07654	finite
2	periodic	1.0	0.765302	finite
2	noise	0.2	-	undefined
2	noise	0.5	-	infinite
2	noise	1.0	3.31381	finite
2	mixed	0.2	-	infinite
2	mixed	0.5	3.17109	finite
2	mixed	1.0	1.75318	finite
3	periodic	0.2	-	infinite
3	periodic	0.5	1.86322	finite
3	periodic	1.0	0.898039	finite
3	noise	0.2	-	undefined
3	noise	0.5	-	infinite
3	noise	1.0	-	infinite
3	mixed	0.2	-	undefined
3	mixed	0.5	-	infinite
3	mixed	1.0	4.71253	finite

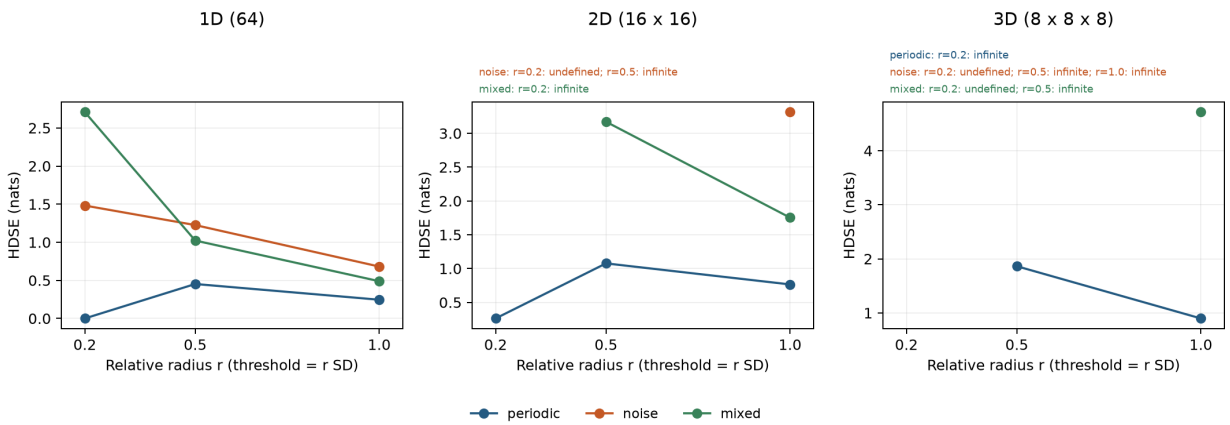


Figure 1. Finite HDSE estimates for the seeded synthetic fields. Text annotations preserve undefined and infinite outcomes; those outcomes are not plotted as zero. Lines connect configured tolerances and serve as visual guides only.

Provenance and reproducibility

Results CSV SHA-256: a4104429035874774f07cb2ec816a1cd656504435f245754d730e3bc06526727

The manifest records hashes and byte counts for generated inputs, strict JSON, the CSV, figure, inspectable Markdown manuscript and PDF, as well as source and configuration inputs. The validator requires the complete artifact set, verifies actual bytes, cross-checks JSON rows against CSV, and inspects PDF text. Re-running the scripts reproduces the specified experiment; a changed configuration or implementation requires regeneration of the manifest and manuscript.

The project adapts docxology/template_code_project at commit 04040a1e8eda0b31ce61db888cd1b3598f1696ca. Its domain-neutral separation of source, tests, scripts, data, manuscript and output is retained. A standalone package, focused synthetic design and reportlab renderer replace the template's monorepo demonstration and publication infrastructure. The supplied copyrighted paper is not included in this repository. No DOI is claimed for this unreleased project.

Limitations and next work

Pair counting remains quadratic in the number of templates. Blocked comparisons bound intermediate memory but do not eliminate that computational growth. High dimension and low tolerance can leave few or no matches, so status and counts must accompany any estimate. Stable ordinal ties are a deterministic convention, not a model of measurement noise. Histogram estimates depend on binning; FFT and SVD estimates depend on preprocessing and normalization. The synthetic experiment does not validate use in ant behavior, biomedical diagnosis, or classification. Independent review, application-specific validation and appropriately controlled scientific studies remain necessary for those uses.

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